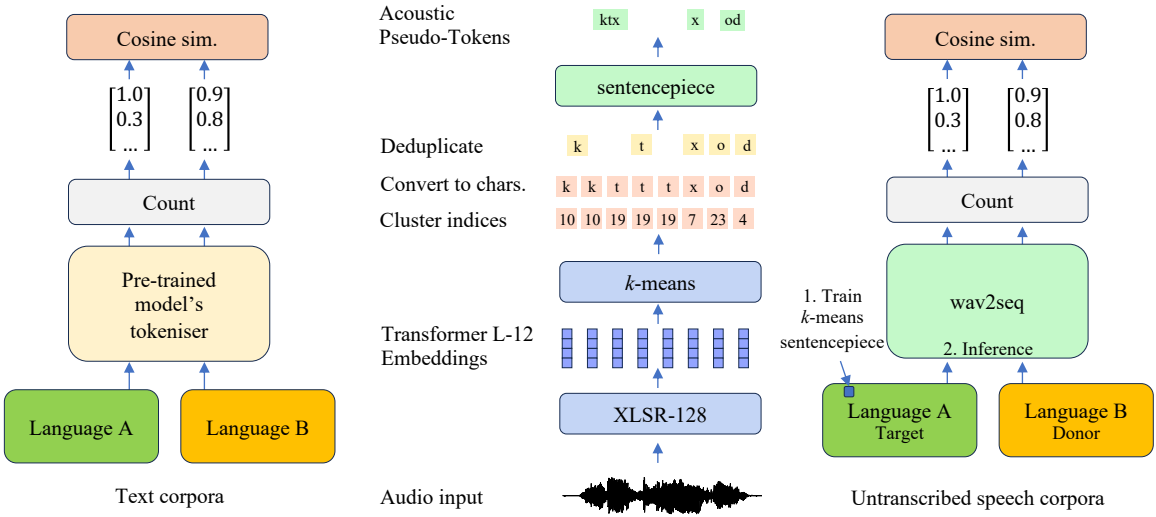
(a) TDS
Cosine sim.
[1.0 0.3 ...]
[0.9 0.8 ...]
Count
Pre-trained model's tokeniser
Language A
Language B
Text corpora
(b) wav2seq
Acoustic Pseudo-Tokens
ktx
x
od
sentencepiece
Deduplicate
k
t
x
o
d
Convert to chars.
k k t t t x o d
Cluster indices
10 10 19 19 19 7 23 4
k-means
Transformer L-12 Embeddings
XLSR-128
Audio input
(c) ATDS
Cosine sim.
[1.0 0.3 ...]
[0.9 0.8 ...]
Count
1. Train k-means sentencepiece
wav2seq
2. Inference
Language A
Target
Language B
Donor
Untranscribed speech corpora